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References

Algina: A comparison of methods for constructing confidence intervals for the squared multiple correlation coefficient **Algina-1999**

James Algina. "A comparison of methods for constructing confidence intervals for the squared multiple correlation coefficient". In: *Multivariate Behavioral Research* 34.4 (Oct. 1999), pp. 493–504. ISSN: 1532-7906. DOI: 10.1207/S15327906MBR3404_5.

Abstract: Four methods for constructing 100(1 - a)% confidence intervals for the population squared multiple correlation coefficient (r2) were compared. In one method the confidence interval was constructed by using the distribution of R^2. Provided r^2 ¿ 0 the coverage probability for this method is exactly 1 - a when the data are multivariate normal. The other three methods are based on results in Olkin and Finn (1995) and are approximate. Results show that each of the approximate methods works very poorly for some combinations of r^2. The method based on the distribution of R^2 is recommended.

Andrews: Heteroskedasticity and autocorrelation consistent covariance matrix estimation **Andrews-1991**

Donald W. K. Andrews. "Heteroskedasticity and autocorrelation consistent covariance matrix estimation". In: *Econometrica* 59.3 (May 1991), p. 817. DOI: 10.2307/2938229.

Abstract: This paper is concerned with the estimation of covariance matrices in the presence of heteroskedasticity and autocorrelation of unknown forms. Currently available estimators that are designed for this context depend upon the choice of a lag truncation parameter and a weighting scheme. Results in the literature provide a condition on the growth rate of the lag truncation pa-

rameter as $T \to \infty$ that is sufficient for consistency. No results are available, however, regarding the choice of lag truncation parameter for a fixed sample size, regarding data-dependent automatic lag truncation parameters, or regarding the choice of weighting scheme. In consequence, available estimators are not entirely operational and the relative merits of the estimators are unknown. This paper addresses these problems. The asymptotic truncated mean squared errors of estimators in a given class are determined and compared. Asymptotically optimal kernel/weighting scheme and bandwidth/lag truncation parameters are obtained using an asymptotic truncated mean squared error criterion. Using these results, data-dependent automatic bandwidth/lag truncation parameters are introduced. The finite sample properties of the estimators are analyzed via Monte Carlo simulation.

Andrews et al.: An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator **Andrews-Monahan-1992**

Donald W. K. Andrews and J. Christopher Monahan. "An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator". In: *Econometrica* 60.4 (July 1992), p. 953. DOI: 10.2307/2951574.

Berkey et al.: Meta-analysis of multiple outcomes by regression with random effects **Berkey-Hoaglin-AntczakBouckoms-etal-1998**

C. S. Berkey et al. "Meta-analysis of multiple outcomes by regression with random effects". In: *Statistics in Medicine* 17.22 (Nov. 1998), pp. 2537–2550. ISSN: 1097-0258. DOI: 10.1002/(sici)1097-0258(19981130)17:22<2537::aid-sim953>3.0.co;2-c.

Abstract: Earlier work showed how to perform fixed-effects meta-analysis of studies or trials when each provides results on more than one outcome per patient and these multiple outcomes are correlated. That fixed-effects generalized-least-squares approach analyzes the multiple outcomes jointly within a single model, and it can include covariates, such as duration of therapy or quality of

trial, that may explain observed heterogeneity of results among the trials. Sometimes the covariates explain all the heterogeneity, and the fixed-effects regression model is appropriate. However, unexplained heterogeneity may often remain, even after taking into account known or suspected covariates. Because fixed-effects models do not make allowance for this remaining unexplained heterogeneity, the potential exists for bias in estimated coefficients, standard errors and p-values. We propose two random-effects approaches for the regression meta-analysis of multiple correlated outcomes. We compare their use with fixed-effects models and with separate-outcomes models in a meta-analysis of periodontal clinical trials. A simulation study shows the advantages of the random-effects approach. These methods also facilitate meta-analysis of trials that compare more than two treatments.

Block et al.: IQ and ego-resiliency: Conceptual and empirical connections and separateness **Block-Kremen-1996**

Jack Block and Adam M. Kremen. "IQ and ego-resiliency: Conceptual and empirical connections and separateness". In: *Journal of Personality and Social Psychology* 70.2 (1996), pp. 349–361. ISSN: 0022-3514. DOI: 10.1037/0022-3514.70.2.349.

Abstract: The constructs of intelligence and ego-resiliency are discussed. The personality implications of "pure intelligence" and "pure ego-resilience" were identified. Intelligence (IQ) was indexed by the Wechsler Adult Intelligence Scale-Revised (WAIS-R) and ego-resiliency by an inventory scale. Residual scores measuring "pure intelligence" and "pure ego-resilience" were correlated with the items of the observer based California Q-sort, used to describe participants. Persons relatively high on ego-resilience tend to be more competent and comfortable in the "fuzzier" interpersonal world; persons defined primarily by raw IQ tend to be effective in the "clearer" world of structured work but tend also to be uneasy with affect and less able to realize satisfying human connections. Gender differences exist in the relations of ego-resilience and intelligence and in their adaptive relevance.

Bolger et al.: Personality and the problems of everyday life: The role of neuroticism in exposure and reactivity to daily stressors **Bolger-Schilling-1991**

Niall Bolger and Elizabeth A. Schilling. "Personality and the problems of everyday life: The role of neuroticism in exposure and reactivity to daily stressors". In: *Journal of Personality* 59.3 (Sept. 1991), pp. 355–386. ISSN: 1467-6494. DOI: 10.1111/j.1467-6494.1991.tb00253.x.

Abstract: This article investigates mechanisms through which neuroticism leads to distress in daily life. Neuroticism may lead to distress through exposing people to a greater number of stressful events, through increasing their reactivity to those events, or through a mechanism unrelated to environmental events. This article evaluates the relative importance of these three explanations. Subjects were 339 persons who provided daily reports of minor stressful events and mood for 6 weeks. Exposure and reactivity to these minor stressors explained over 40% of the distress difference between high- and low-neuroticism subjects. Reactivity to stressors accounted for twice as much of the distress difference as exposure to stressors. These results suggest that reactions within stressful situations are more important than situation selection in explaining how neuroticism leads to distress in daily life.

Bolger et al.: A framework for studying personality in the stress process **Bolger-Zuckerman-1995**

Niall Bolger and Adam Zuckerman. "A framework for studying personality in the stress process". In: *Journal of Personality and Social Psychology* 69.5 (1995), pp. 890–902. ISSN: 0022-3514. DOI: 10.1037/0022-3514.69.5.890.

Abstract: This article presents a framework for studying personality in the stress process. The framework specifies that personality may affect both exposure and reactivity to stressful events and that both processes may explain how personality affects health and psychological outcomes. The framework also specifies that personality differences in reactivity may be due to differential choice of coping efforts and differential effectiveness of those efforts. In a 14-day daily diary study

of 94 students, this framework was used to analyze the links among neuroticism, daily interpersonal conflicts, coping with conflicts, and distress. Results showed that high-neuroticism participants had greater exposure and reactivity to conflicts. Furthermore, high- and low-neuroticism participants differed both in their choice of coping efforts and in the effectiveness of those efforts, a possibility not considered in previous models of personality in the stress process.

Bollen et al.: Direct and indirect effects: Classical and bootstrap estimates of variability **Bollen-Stine-1990**

Kenneth A. Bollen and Robert Stine. "Direct and indirect effects: Classical and bootstrap estimates of variability". In: *Sociological Methodology* 20 (1990), p. 115. DOI: 10.2307/271084.

Abstract: The decomposition of effects in structural equation models has been of considerable interest to social scientists. Finite-sample or asymptotic results for the sampling distribution of estimators of direct effects are widely available. Statistical inferences about indirect effects have relied exclusively on asymptotic methods which assume that the limiting distribution of the estimator is normal, with a standard error derived from the delta method. We examine bootstrap procedures as another way to generate standard errors and confidence intervals and to estimate the sampling distributions of estimators of direct and indirect effects. We illustrate the classical and the bootstrap methods with three empirical examples. We find that in a moderately large sample, the bootstrap distribution of an estimator is close to that assumed with the classical and delta methods but that in small samples, there are some differences. Bootstrap methods provide a check on the classical and delta methods when the latter are applied under less than ideal conditions.

Celeux et al.: An entropy criterion for assessing the number of clusters in a mixture model **Celeux-Soromenho-1996**

Gilles Celeux and Gilda Soromenho. "An entropy criterion for assessing the number of clusters in a

mixture model". In: *Journal of Classification* 13.2 (Sept. 1996), pp. 195–212. ISSN: 1432-1343. DOI: 10.1007/bf01246098.

Abstract: In this paper, we consider an entropy criterion to estimate the number of clusters arising from a mixture model. This criterion is derived from a relation linking the likelihood and the classification likelihood of a mixture. Its performance is investigated through Monte Carlo experiments, and it shows favorable results compared to other classical criteria.

Clark et al.: Tripartite model of anxiety and depression: Psychometric evidence and taxonomic implications **Clark-Watson-1991**

Lee Anna Clark and David Watson. "Tripartite model of anxiety and depression: Psychometric evidence and taxonomic implications". In: *Journal of Abnormal Psychology* 100.3 (Aug. 1991), pp. 316–336. ISSN: 0021-843X. DOI: 10.1037/0021-843x.100.3.316.

Abstract: Reviews psychometric and other evidence relevant to mixed anxiety-depression. Properties of anxiety and depression measures, including the convergent and discriminant validity of self- and clinical ratings, and interrater reliability, are examined in patient and normal samples. Results suggest that anxiety and depression can be reliably and validly assessed; moreover, although these disorders share a substantial component of general affective distress, they can be differentiated on the basis of factors specific to each syndrome. L. A. Clark and D. Watson also review evidence for these specific factors, examining the influence of context and scale content on ratings, factor analytic studies, and the role of low positive affect in depression. With these data, Clark and Watson argue for a tripartite structure consisting of general distress, physiological hyperarousal (specific anxiety), and anhedonia (specific depression), and they propose a diagnosis of mixed anxiety-depression.

Collins et al.: Latent class models for stage-sequential dynamic latent variables **Collins-Wugalter-1992**

Linda M. Collins and Stuart E. Wugalter. "Latent class models for stage-sequential dynamic latent

variables". In: *Multivariate Behavioral Research* 27.1 (Jan. 1992), pp. 131–157. ISSN: 1532-7906. DOI: 10.1207/s15327906mbr2701_8.

Abstract: Stage-sequential dynamic latent variables are of interest in many longitudinal studies. Measurement theory for these latent variables, called Latent Transition Analysis (LTA), can be found in recent generalizations of latent class theory. LTA expands the latent Markov model to allow applications to more complex latent variables and the use of multiple indicators. Because complex latent class models result in sparse contingency tables, that may lead to poor parameter estimation, a simulation study was conducted in order to determine whether model parameters are recovered adequately by LTA, and whether additional indicators result in better measurement or in impossibly sparse tables. The results indicated that parameter recovery was satisfactory overall, although as expected the standard errors were large in some conditions with few subjects. The simulation also indicated that at least within the conditions examined here, the benefits of adding indicators outweigh the costs. Additional indicators improved standard errors, even in conditions producing extremely sparse tables. An example of LTA analysis of empirical data on math skill development is presented.

M. L. Cooper et al.: Drinking to regulate positive and negative emotions: A motivational model of alcohol use **Cooper-Frone-Russell-etal-1995**

M. Lynne Cooper et al. "Drinking to regulate positive and negative emotions: A motivational model of alcohol use". In: *Journal of Personality and Social Psychology* 69.5 (Nov. 1995), pp. 990–1005. ISSN: 0022-3514. DOI: 10.1037/0022-3514.69.5.990.

Abstract: The present study proposed and tested a motivational model of alcohol use in which people are hypothesized to use alcohol to regulate both positive and negative emotions. Two central premises underpin this model: (a) that enhancement and coping motives for alcohol use are proximal determinants of alcohol use and abuse through which the influence of expectancies, emotions, and other individual differences are mediated and (b) that enhancement and coping motives

represent phenomenologically distinct behaviors having both unique antecedents and consequences. This model was tested in 2 random samples (1 of adults, 1 of adolescents) using a combination of moderated regression and path analysis corrected for measurement error. Results revealed strong support for the hypothesized model in both samples and indicate the importance of distinguishing psychological motives for alcohol use.

DeNeve et al.: The happy personality: A meta-analysis of 137 personality traits and subjective well-being **DeNeve-Cooper-1998**

Kristina M. DeNeve and Harris Cooper. "The happy personality: A meta-analysis of 137 personality traits and subjective well-being". In: *Psychological Bulletin* 124.2 (1998), pp. 197–229. ISSN: 0033-2909. DOI: 10.1037/0033-2909.124.2.197.

Abstract: This meta-analysis used 9 literature search strategies to examine 137 distinct personality constructs as correlates of subjective well-being (SWB). Personality was found to be equally predictive of life satisfaction, happiness, and positive affect, but significantly less predictive of negative affect. The traits most closely associated with SWB were repressive-defensiveness, trust, emotional stability, locus of control-chance, desire for control, hardiness, positive affectivity, private collective self-esteem, and tension. When personality traits were grouped according to the Big Five factors, Neuroticism was the strongest predictor of life satisfaction, happiness, and negative affect. Positive affect was predicted equally well by Extraversion and Agreeableness. The relative importance of personality for predicting SWB, how personality might influence SWB, and limitations of the present review are discussed.

Dishion et al.: Parental monitoring and the prevention of child and adolescent problem behavior: A conceptual and empirical formulation **Dishion-McMahon-1998**

Thomas J. Dishion and Robert J. McMahon. "Parental monitoring and the prevention of child and

adolescent problem behavior: A conceptual and empirical formulation". In: *Clinical Child and Family Psychology Review* 1.1 (Mar. 1998), pp. 61–75. ISSN: 1573-2827. DOI: 10.1023/a:1021800432380.

Abstract: The present report accomplishes three goals. First, to provide an empirical rationale for placing parental monitoring of children's adaptations as a key construct in development and prevention research. Second, to stimulate more research on parental monitoring and provide an integrative framework for various research traditions as well as developmental periods of interest. Third, to discuss current methodological issues that are developmentally and culturally sensitive and based on sound measurement. Possible intervention and prevention strategies that specifically target parental monitoring are discussed.

Dumenci et al.: A latent trait-state model of adolescent depression using the Center for Epidemiologic Studies-Depression Scale **Dumenci-Windle-1996**

Levent Dumenci and Michael Windle. "A latent trait-state model of adolescent depression using the Center for Epidemiologic Studies-Depression Scale". In: *Multivariate Behavioral Research* 31.3 (July 1996), pp. 313–330. ISSN: 1532-7906. DOI: 10.1207/s15327906mbr3103_3.

Abstract: Utilized the latent trait-state model for estimating stable and changing components of depressive symptomology in adolescents. The factorial structure of the Center for Epidemiologic Studies-Depression Scale (CES-D) was assessed separately for males and females at 4 measurement occasions, at 6-mo intervals. The variance decomposition of general trait, state, specific trait, and random error parameters for the CES-D scores was estimated simultaneously and tested statistically. Parameter estimates indicated that the CES-D measured both trait- and state-depression about equally well, and that the trait-specific variance parameter was statistically significant, but substantially smaller than those associated with general trait- and state-depression. Findings are discussed with regard to depressive mood fluctuations among adolescents and the potential usefulness of the latent trait-state model to capture such dynamic features of development.

Eilers et al.: Flexible smoothing with B-splines and penalties **Eilers-Marx-1996**

Paul H. C. Eilers and Brian D. Marx. "Flexible smoothing with B-splines and penalties". In: *Statistical Science* 11.2 (May 1996). ISSN: 0883-4237. DOI: 10.1214/ss/1038425655.

Abstract: B-splines are attractive for nonparametric modelling, but choosing the optimal number and positions of knots is a complex task. Equidistant knots can be used, but their small and discrete number allows only limited control over smoothness and fit. We propose to use a relatively large number of knots and a difference penalty on coefficients of adjacent B-splines. We show connections to the familiar spline penalty on the integral of the squared second derivative. A short overview of B-splines, of their construction and of penalized likelihood is presented. We discuss properties of penalized B-splines and propose various criteria for the choice of an optimal penalty parameter. Nonparametric logistic regression, density estimation and scatterplot smoothing are used as examples. Some details of the computations are presented.

Hoogland et al.: Robustness studies in covariance structure modeling: An overview and a meta-analysis **Hoogland-Boomsma-1998**

Jeffrey J. Hoogland and Anne Boomsma. "Robustness studies in covariance structure modeling: An overview and a meta-analysis". In: *Sociological Methods & Research* 26.3 (Feb. 1998), pp. 329–367. ISSN: 1552-8294. DOI: 10.1177/0049124198026003003.

Abstract: In covariance structure modeling several estimation methods are available. The robustness of an estimator against specific violations of assumptions can be determined empirically by means of a Monte Carlo study. Many such studies in covariance structure analysis have been published, but the conclusions frequently seem to contradict each other An overview of robustness studies in covariance structure analysis is given, and an attempt is made to generalize their findings. Robustness studies are described and distinguished from each other systematically by means of certain characteristics. These characteristics serve as explanatory variables in a meta-analysis concerning

the behavior of parameter estimators, standard error estimators, and goodness-of-fit statistics when the model is correctly specified.

Kenny et al.: The trait-state-error model for multiwave data **Kenny-Zautra-1995**

David A. Kenny and Alex Zautra. "The trait-state-error model for multiwave data". In: *Journal of Consulting and Clinical Psychology* 63.1 (1995), pp. 52–59. ISSN: 0022-006X. DOI: 10.1037/0022-006x.63.1.52.

Abstract: Although researchers in clinical psychology routinely gather data in which many individuals respond at multiple times, there is not a standard way to analyze such data. A new approach for the analysis of such data is described. It is proposed that a person's current standing on a variable is caused by 3 sources of variance: a term that does not change (trait), a term that changes (state), and a random term (error). It is shown how structural equation modeling can be used to estimate such a model. An extended example is presented in which the correlations between variables are quite different at the trait, state, and error levels.

Li et al.: Large-sample significance levels from multiply imputed data using moment-based statistics and an F reference distribution **Li-Raghunathan-Rubin-1991**

K. H. Li, Trivellore Eachambadi Raghunathan, and Donald B. Rubin. "Large-sample significance levels from multiply imputed data using moment-based statistics and an F reference distribution". In: *Journal of the American Statistical Association* 86.416 (Dec. 1991), pp. 1065–1073. DOI: 10.1080/01621459.1991.10475152.

Abstract: We present a procedure for computing significance levels from data sets whose missing values have been multiply imputed data. This procedure uses moment-based statistics, $m \leq 3$ repeated imputations, and an F reference distribution. When $m = \infty$, we show first that our procedure is essentially the same as the ideal procedure in cases of practical importance and, second, that its deviations from the ideal are basically a function of the coefficient of variation of

the canonical ratios of complete to observed information. For small m our procedure's performance is largely governed by this coefficient of variation and the mean of these ratios. Using simulation techniques with small m, we compare our procedure's actual and nominal large-sample significance levels and conclude that it is essentially calibrated and thus represents a definite improvement over previously available procedures. Furthermore, we compare the large-sample power of the procedure as a function of m and other factors, such as the dimensionality of the estimand and fraction of missing information, to provide guidance on the choice of the number of imputations; generally, we find the loss of power due to small m to be quite modest in cases likely to occur in practice.

Lyapunov: The general problem of the stability of motion **Lyapunov-1992**

Alexandr Mikhailovich Lyapunov. "The general problem of the stability of motion". In: *International Journal of Control* 55.3 (Mar. 1992), pp. 531–534. ISSN: 1366-5820. DOI: 10.1080/00207179208934253.

Lykken et al.: Happiness is a stochastic phenomenon **Lykken-Tellegen-1996**

David Lykken and Auke Tellegen. "Happiness is a stochastic phenomenon". In: *Psychological Science* 7.3 (May 1, 1996), pp. 186–189. ISSN: 1467-9280. DOI: 10.1111/j.1467-9280.1996.tb00355.x.

Abstract: Happiness, or subjective well-being, was measured on a birth-record-based sample of several thousand middle-aged twins using the Well-Being (WB) scale of the Multidimensional Personality Questionnaire. Neither socioeconomic status, educational attainment, family income, marital status, nor an indicant of religious commitment could account for more than about 3% of the variance in WB. From 44% to 52% of the variance in WB, however, is associated with genetic variation. Based on the retest of smaller samples of twins after intervals of 4.5 and 10 years, we estimate that the heritability of the stable component of subjective well-being approaches 80%.

MacKinnon: Analysis of mediating variables in prevention and intervention research

MacKinnon-1994

David P. MacKinnon. "Analysis of mediating variables in prevention and intervention research". In: *NIDA research monograph* 139 (1994), pp. 127–153.

Abstract: Mediational analysis is one way to test specific hypotheses derived from theory. Although this analysis has been suggested in the prevention literature, mediation analysis rarely is conducted. As the field of prevention matures, more questions about how prevention programs work (or fail to work) will emerge. Studies of mediation can address these questions, thereby reducing the cost and enhancing the impact of prevention programs. The methods outlined here can be applied in the evaluation of primary, secondary, and tertiary prevention programs. Since most prevention studies include measurement of some mediating constructs, mediation effects can be assessed on many existing data sets. Mediation analysis can be used to test ideas about prevention.

MacKinnon et al.: Estimating mediated effects in prevention studies

MacKinnon-Dwyer-1993

David P. MacKinnon and James H. Dwyer. "Estimating mediated effects in prevention studies". In: *Evaluation Review* 17.2 (Apr. 1993), pp. 144–158. DOI: 10.1177/0193841x9301700202.

Abstract: The purpose of this article is to describe statistical procedures to assess how prevention and intervention programs achieve their effects. The analyses require the measurement of intervening or mediating variables hypothesized to represent the causal mechanism by which the prevention program achieves its effects. Methods to estimate mediation are illustrated in the evaluation of a health promotion program designed to reduce dietary cholesterol and a school-based drug prevention program. The methods are relatively easy to apply and the information gained from such analyses should add to our understanding of prevention.

Muthén et al.: General longitudinal modeling of individual differences in experimental designs: A latent variable framework for analysis and power estimation

Muthen-Curran-1997

Bengt O. Muthén and Patrick J. Curran. "General longitudinal modeling of individual differences in experimental designs: A latent variable framework for analysis and power estimation". In: *Psychological Methods* 2.4 (Dec. 1997), pp. 371–402. DOI: 10.1037/1082-989x.2.4.371.

Abstract: The generality of latent variable modeling of individual differences in development over time is demonstrated with a particular emphasis on randomized intervention studies. First, a brief overview is given of biostatistical and psychometric approaches to repeated measures analysis. Second, the generality of the psychometric approach is indicated by some nonstandard models. Third, a multiple-population analysis approach is proposed for the estimation of treatment effects. The approach clearly describes the treatment effect as development that differs from normative, control-group development. This framework allows for interactions between treatment and initial status in their effects on development. Finally, an approach for the estimation of power to detect treatment effects in this framework is demonstrated. Illustrations of power calculations are carried out with artificial data, varying the sample sizes, number of timepoints, and treatment effect sizes. Real data are used to illustrate analysis strategies and power calculations. Further modeling extensions are discussed.

Muthén et al.: Finite mixture modeling with mixture outcomes using the EM algorithm

Muthen-Shedden-1999

Bengt O. Muthén and Kerby Shedden. "Finite mixture modeling with mixture outcomes using the EM algorithm". In: *Biometrics* 55.2 (June 1999), pp. 463–469. ISSN: 1541-0420. DOI: 10.1111/j.0006-341x.1999.00463.x.

Abstract: This paper discusses the analysis of an extended finite mixture model where the latent classes corresponding to the mixture components for one set of observed variables influence a second

set of observed variables. The research is motivated by a repeated measurement study using a random coefficient model to assess the influence of latent growth trajectory class membership on the probability of a binary disease outcome. More generally, this model can be seen as a combination of latent class modeling and conventional mixture modeling. The EM algorithm is used for estimation. As an illustration, a random-coefficient growth model for the prediction of alcohol dependence from three latent classes of heavy alcohol use trajectories among young adults is analyzed.

Oehlert: A note on the delta method **Oehlert-1992**

Gary W. Oehlert. "A note on the delta method". In: *The American Statistician* 46.1 (Feb. 1992), pp. 27–29. ISSN: 1537-2731. DOI: 10.1080/00031305.1992.10475842.

Abstract: The delta method is an intuitive technique for approximating the moments of functions of random variables. This note reviews the delta method and conditions under which delta-method approximate moments are accurate.

Olkin et al.: Correlations redux **Olkin-Finn-1995**

Ingram Olkin and Jeremy D. Finn. "Correlations redux". In: *Psychological Bulletin* 118.1 (July 1995), pp. 155–164. ISSN: 0033-2909. DOI: 10.1037/0033-2909.118.1.155.

Oud et al.: Longitudinal factor score estimation using the Kalman filter **Oud-vandenBercken-Essers-1990**

Johan H. Oud, John H. van den Bercken, and Raymond J. Essers. "Longitudinal factor score estimation using the Kalman filter". In: *Applied Psychological Measurement* 14.4 (Dec. 1990), pp. 395–418. DOI: 10.1177/014662169001400406.

Abstract: The advantages of the Kalman filter as a factor score estimator in the presence of longitudinal data are described. Because the Kalman filter presupposes the availability of a dynamic

state space model, the state space model is reviewed first, and it is shown to be translatable into the LISREL model. Several extensions of the LISREL model specification are discussed in order to enhance the applicability of the Kalman filter for behavioral research data. The Kalman filter and its main properties are summarized. Relationships are shown between the Kalman filter and two well-known cross-sectional factor score estimators: the regression estimator, and the Bartlett estimator. The indeterminacy problem of factor scores is also discussed in the context of Kalman filtering, and the differences are described between Kalman filtering on the basis of a zero-means and a structured-means LISREL model. By using a structured-means LISREL model, the Kalman filter is capable of estimating absolute latent developmental curves. An educational research example is presented. Index terms: factor score estimation, indeterminacy of factor scores, Kalman filter, L,ISREL longitudinal LISREL modeling, longitudinal factor analysis, state space modeling.

Politis et al.: The stationary bootstrap **Politis-Romano-1994**

Dimitris N. Politis and Joseph P. Romano. "The stationary bootstrap". In: *Journal of the American Statistical Association* 89.428 (Dec. 1994), pp. 1303–1313. ISSN: 1537-274X. DOI: 10.1080/01621459.1994.10476870.

Abstract: This article introduces a resampling procedure called the stationary bootstrap as a means of calculating standard errors of estimators and constructing confidence regions for parameters based on weakly dependent stationary observations. Previously, a technique based on resampling blocks of consecutive observations was introduced to construct confidence intervals for a parameter of the m-dimensional joint distribution of m consecutive observations, where m is fixed. This procedure has been generalized by constructing a "blocks of blocks" resampling scheme that yields asymptotically valid procedures even for a multivariate parameter of the whole (i.e., infinite-dimensional) joint distribution of the stationary sequence of observations. These methods share the construction of resampling blocks of observations to form a pseudo-time series, so that the statistic of interest may be recalculated based on the resampled data set. But in the context of applying this method to stationary data, it is natural to require the resampled pseudo-time series to be stationary (condi-

tional on the original data) as well. Although the aforementioned procedures lack this property, the stationary procedure developed here is indeed stationary and possesses other desirable properties. The stationary procedure is based on resampling blocks of random length, where the length of each block has a geometric distribution. In this article, fundamental consistency and weak convergence properties of the stationary resampling scheme are developed.

Robey et al.: Type I error and the number of iterations in Monte Carlo studies of robustness **Robey-Barcikowski-1992**

Randall R. Robey and Robert S. Barcikowski. "Type I error and the number of iterations in Monte Carlo studies of robustness". In: *British Journal of Mathematical and Statistical Psychology* 45.2 (Nov. 1992), pp. 283–288. DOI: 10.1111/j.2044-8317.1992.tb00993.x.

Abstract: A recent survey of simulation studies concluded that an overwhelming majority of papers do not report a rationale for the decision regarding the number of Monte Carlo iterations. A surprisingly large number of reports do not contain a justifiable definition of robustness and many studies are conducted with an insufficient number of iterations to achieve satisfactory statistical conclusion validity. The implication is that we do not follow our own advice regarding the management of Type I and Type II errors when conducting Monte Carlo experiments. This paper reports a straightforward application of a well-known procedure for the purpose of objectively determining the exact number of iterations necessary to confidently detect departures from robustness in Monte Carlo results. A table of the number of iterations necessary to detect departures from a series of nominal Type I error rates is included.

Robins et al.: Identifiability and exchangeability for direct and indirect effects **Robins-Greenland-1992**

James M. Robins and Sander Greenland. "Identifiability and exchangeability for direct and indirect

effects". In: *Epidemiology* 3.2 (Mar. 1992), pp. 143–155. ISSN: 1044-3983. DOI: 10.1097/00001648-199203000-00013.

Abstract: We consider the problem of separating the direct effects of an exposure from effects relayed through an intermediate variable (indirect effects). We show that adjustment for the intermediate variable, which is the most common method of estimating direct effects, can be biased. We also show that, even in a randomized crossover trial of exposure, direct and indirect effects cannot be separated without special assumptions; in other words, direct and indirect effects are not separately identifiable when only exposure is randomized. If the exposure and intermediate never interact to cause disease and if intermediate effects can be controlled, that is, blocked by a suitable intervention, then a trial randomizing both exposure and the intervention can separate direct from indirect effects. Nonetheless, the estimation must be carried out using the G-computation algorithm. Conventional adjustment methods remain biased. When exposure and the intermediate interact to cause disease, direct and indirect effects will not be separable even in a trial in which both the exposure and the intervention blocking intermediate effects are randomly assigned. Nonetheless, in such a trial, one can still estimate the fraction of exposure-induced disease that could be prevented by control of the intermediate. Even in the absence of an intervention blocking the intermediate effect, the fraction of exposure-induced disease that could be prevented by control of the intermediate can be estimated with the G-computation algorithm if data are obtained on additional confounding variables.

Saunders et al.: Development of the Alcohol Use Disorders Identification Test (AUDIT): WHO Collaborative Project on Early Detection of Persons with Harmful Alcohol Consumption-II **Saunders-Assland-Babor-etal-1993**

John B. Saunders et al. "Development of the Alcohol Use Disorders Identification Test (AUDIT): WHO Collaborative Project on Early Detection of Persons with Harmful Alcohol Consumption-II". In: *Addiction* 88.6 (June 1993), pp. 791–804. ISSN: 1360-0443. DOI: 10.1111/j.1360-0443.1993.tb02093.x.

Abstract: The Alcohol Use Disorders Identification Test (AUDIT) has been developed from a six-country WHO collaborative project as a screening instrument for hazardous and harmful alcohol consumption. It is a 10-item questionnaire which covers the domains of alcohol consumption, drinking behaviour, and alcohol-related problems. Questions were selected from a 150-item assessment schedule (which was administered to 1888 persons attending representative primary health care facilities) on the basis of their representativeness for these conceptual domains and their perceived usefulness for intervention. Responses to each question are scored from 0 to 4, giving a maximum possible score of 40. Among those diagnosed as having hazardous or harmful alcohol use, 92% had an AUDIT score of 8 or more, and 94% of those with non-hazardous consumption had a score of less than 8. AUDIT provides a simple method of early detection of hazardous and harmful alcohol use in primary health care settings and is the first instrument of its type to be derived on the basis of a cross-national study.

Schulenberg et al.: Getting drunk and growing up: trajectories of frequent binge drinking during the transition to young adulthood

Schulenberg-OMalley-Bachman-etal-1996

John E. Schulenberg et al. "Getting drunk and growing up: trajectories of frequent binge drinking during the transition to young adulthood". In: *Journal of Studies on Alcohol* 57.3 (May 1996), pp. 289–304. ISSN: 1934-2683. DOI: 10.15288/jsa.1996.57.289.

Abstract: Objective: The purpose of this study was: (1) to identify different trajectories of frequent binge drinking during the transition to young adulthood; (2) to validate the trajectories by relating them to behaviors and attitudes concerning alcohol and other drug use; and (3) to distinguish among the trajectories according to demographic characteristics and lifestyle experiences typical of the transition to young adulthood. Method: Four waves of national panel data were obtained from the Monitoring the Future project; 9,945 weighted cases from the 1976-85 high school senior year cohorts were surveyed at biennial intervals between ages 18 and 24. Frequent binge drinking was defined as having five or more drinks in a row at least twice in the past two weeks. Results: Six

distinct frequent binge drinking trajectory groups were specified a priori and confirmed with cluster analysis: Never, Rare, Chronic, Decreased, Increased and "Fling." Repeated measures ANOVAS revealed that the trajectories corresponded to patterns of change and stability in problems with alcohol, attitudes about heavy drinking, peer heavy drinking and illicit drug use. Results from logistic regression analyses predicting diverging and converging trajectories provided some support for the general hypothesis that trajectories of Chronic and Increased frequent binge drinking over time are associated with difficulties in negotiating the transition to young adulthood. Conclusions: The findings provide strong evidence for wide developmental variation in drinking patterns in the population, variation that is obscured by more aggregate-level considerations. The developmental variation in frequent binge drinking during the transition to young adulthood reflects systematic variation in success and difficulties with negotiating the transition.

Shapiro et al.: On the treatment of correlation structures as covariance structures **Shapiro-Browne-1990**

A. Shapiro and M.W. Browne. "On the treatment of correlation structures as covariance structures". In: *Linear Algebra and its Applications* 127 (1990), pp. 567–587. ISSN: 0024-3795. DOI: 10.1016/0024-3795(90)90362-g.

Abstract: Necessary and sufficient conditions are provided for minimum discrepancy methods, intended for covariance structures, to retain their asymptotic properties in the analysis of correlation structures. Examples of correlation structures satisfying these conditions are considered, and alternative discrepancy functions, which are always appropriate for correlation structures under normality assumptions, are discussed.

Stoffer et al.: Bootstrapping state-space models: Gaussian maximum likelihood estimation and the Kalman filter **Stoffer-Wall-1991**

David S. Stoffer and Kent D. Wall. "Bootstrapping state-space models: Gaussian maximum likeli-

hood estimation and the Kalman filter". In: *Journal of the American Statistical Association* 86.416 (Dec. 1991), pp. 1024–1033. DOI: 10.1080/01621459.1991.10475148.

Abstract: The bootstrap is proposed as a method for assessing the precision of Gaussian maximum likelihood estimates of the parameters of linear state-space models. Our results also apply to autoregressive moving average models, since they are a special case of state-space models. It is shown that for a time-invariant, stable system, the bootstrap applied to the innovations yields asymptotically consistent standard errors. To investigate the performance of the bootstrap for finite sample lengths, simulation results are presented for a two-state model with 50 and 100 observations; two cases are investigated, one with real characteristic roots and one with complex characteristic roots. The bootstrap is then applied to two real data sets, one used in a test for efficient capital markets and one used to develop an autoregressive integrated moving average model for quarterly earnings data. We find the bootstrap to be of definite value over the conventional asymptotics.

Stram et al.: Variance components testing in the longitudinal mixed effects model **Stram-Lee-1994**

Daniel O. Stram and Jae Won Lee. "Variance components testing in the longitudinal mixed effects model". In: *Biometrics* 50.4 (Dec. 1994), pp. 1171–1177. ISSN: 1541-0420. DOI: 10.2307/2533455.

Suls et al.: Emotional reactivity to everyday problems, affective inertia, and neuroticism **Suls-Green-Hillis-1998**

Jerry Suls, Peter Green, and Stephen Hillis. "Emotional reactivity to everyday problems, affective inertia, and neuroticism". In: *Personality and Social Psychology Bulletin* 24.2 (Feb. 1998), pp. 127–136. ISSN: 1552-7433. DOI: 10.1177/0146167298242002.

Abstract: A naturalistic diary recording study was conducted to assess affective responses to everyday stress. Community-residing male participants made diary recordings regarding problem occurrence and mood several times a day for 8 days. In addition to reporting more frequent daily

problems, persons scoring high in neuroticism were more reactive to stressors and were more distressed by recurrent problems than were persons scoring low in neuroticism. New problems affected everyone comparably. There was also evidence of affective inertia, such that bad mood was more likely to carry over to the next assessment. This lag effect tended to be stronger among more neurotic individuals.

Tibshirani: Regression shrinkage and selection via the lasso **Tibshirani-1996**

Robert J. Tibshirani. "Regression shrinkage and selection via the lasso". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 58.1 (Jan. 1996), pp. 267–288. ISSN: 1467-9868. DOI: 10.1111/j.2517-6161.1996.tb02080.x.

Abstract: We propose a new method for estimation in linear models. The 'lasso' minimizes the residual sum of squares subject to the sum of the absolute value of the coefficients being less than a constant. Because of the nature of this constraint it tends to produce some coefficients that are exactly 0 and hence gives interpretable models. Our simulation studies suggest that the lasso enjoys some of the favourable properties of both subset selection and ridge regression. It produces interpretable models like subset selection and exhibits the stability of ridge regression. There is also an interesting relationship with recent work in adaptive function estimation by Donoho and Johnstone. The lasso idea is quite general and can be applied in a variety of statistical models: extensions to generalized regression models and tree-based models are briefly described.

van Houwelingen et al.: A bivariate approach to meta-analysis

vanHouwelingen-Zwinderman-Stijnen-1993

Hans C. van Houwelingen, Koos H. Zwinderman, and Theo Stijnen. "A bivariate approach to meta-analysis". In: *Statistics in Medicine* 12.24 (Dec. 1993), pp. 2273–2284. ISSN: 1097-0258. DOI: 10.1002/sim.4780122405.

Abstract: The usual meta-analysis of a sequence of randomized clinical trials only considers the difference between two treatments and produces a point estimate and a confidence interval for a parameter that measures this difference. The usual parameter is the log(odds ratio) linked to Mantel-Haenszel methodology. Inference is made either under the assumption of homogeneity or in a random effects model that takes account of heterogeneity between trials. This paper has two goals. The first is to present a likelihood based method for the estimation of the parameters in the random effects model, which avoids the use of approximating Normal distributions. The second goal is to extend this method to a bivariate random effects model, in which the effects in both groups are supposed random. In this way inference can be made about the relationship between improvement and baseline effect. The method is demonstrated by a meta-analysis dataset of Collins and Langman.

Von Korff et al.: The relationship between pain and depression VonKorff-Simon-1996

Michael Von Korff and Gregory Simon. "The relationship between pain and depression". In: *British Journal of Psychiatry* 168.S30 (June 1996), pp. 101–108. ISSN: 1472-1465. DOI: 10.1192/s0007125000298474.

Abstract: Empirical results from epidemiological studies on pain-depression comorbidity in primary care and population samples have shown that: (a) pain is as strongly associated with anxiety as with depressive disorders; (b) characteristics that most strongly predict depression are diffuseness of pain and the extent to which pain interferes with activities; (c) certain psychological symptoms (low energy, disturbed sleep, worry) are prominent among pain patients, while others (guilt, loneliness) are not; (d) depression and pain dysfunction are evident early in the natural history of pain, but dysfunction and distress are often transient; and (e) among initially dysfunctional pain patients whose dysfunction is chronic, depression levels do not improve but neither do they increase over time with chronicity alone. These results seem consistent with these mechanisms of pain-depression comorbidity; (1) a trait of susceptibility to both dysphoric physical symptoms (including pain) and psychological symptoms (including depression), and a state of somatosensory amplification in which psychological distress amplifies dysphoric physical sensations (including pain); (2) psychological

illness and behavioural dysfunction being interrelated features of a maladaptive response to pain evident early in the natural history of the condition, and often resolving during an early recovery phase; (3) pain constituting a significant physical and psychological stressor that may induce or exacerbate psychological distress. Thus, pain and psychological illness should be viewed as having reciprocal psychological and behavioural effects involving both processes of illness expression and adaption, as well as pain having specific effects on emotional state and behavioural function.

Wechsler et al.: A gender-specific measure of binge drinking among college students

Wechsler-Dowdall-Davenport-etal-1995

Henry Wechsler et al. "A gender-specific measure of binge drinking among college students". In: *American Journal of Public Health* 85.7 (July 1995), pp. 982–985. ISSN: 1541-0048. DOI: 10.2105/ajph.85.7.982.

Abstract: This study examined the relationship of volume of alcohol consumed to the occurrence of alcohol-related problems among male and female college students to develop a gender-specific measure of heavy episodic or binge drinking by college students for public health research. A self-administered survey was mailed to a national representative sample of students at 140 4-year colleges in 40 states and the District of Columbia. A total of 17,592 college students completed the survey. Women who typically drink four drinks in a row were found to have roughly the same likelihood of experiencing drinking-related problems as men who typically drink five drinks in a row. Use of the same standard for both sexes underestimates binge drinking and the negative health risks for women.

Wilkinson et al.: Statistical methods in psychology journals: Guidelines and explanations **Wilkinson-TaskForceonStatisticalInference-1999**

Leland Wilkinson and Task Force on Statistical Inference. "Statistical methods in psychology journals: Guidelines and explanations". In: *American Psychologist* 54.8 (Aug. 1999), pp. 594–604. ISSN: 0003-066X. DOI: 10.1037/0003-066x.54.8.594.